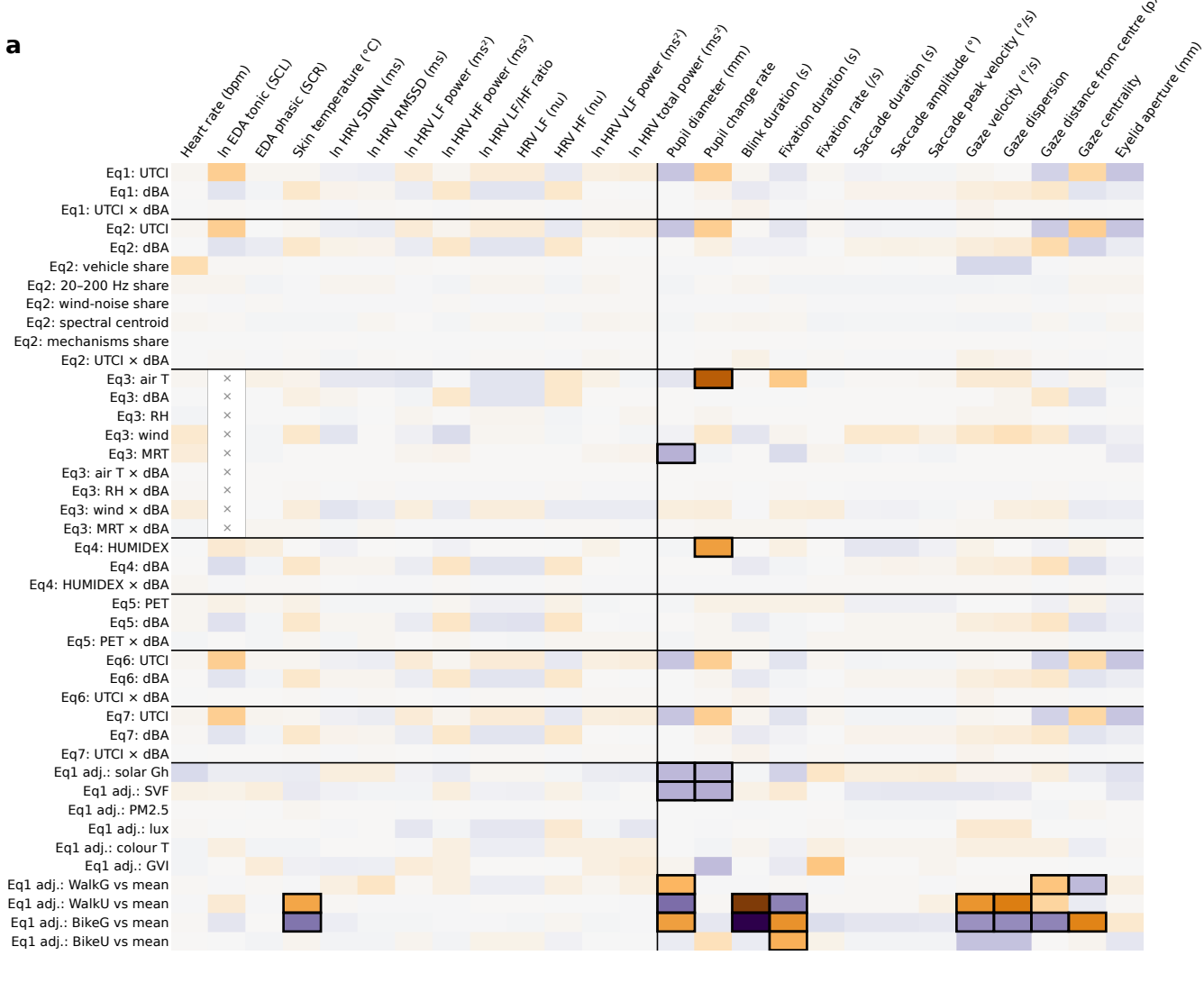
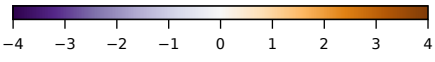




sign(β) × −log10(q) (black box: q < 0.05; ×: not estimable)

Eq1–Eq7 on 10 s time series (bout-centred exposures, AR(1), CR2).
q = BH per term within equation across 26 outcomes.
Exposure-term tests with q < 0.05: 3 of 832. Eq3 ln EDA tonic: model not estimable.

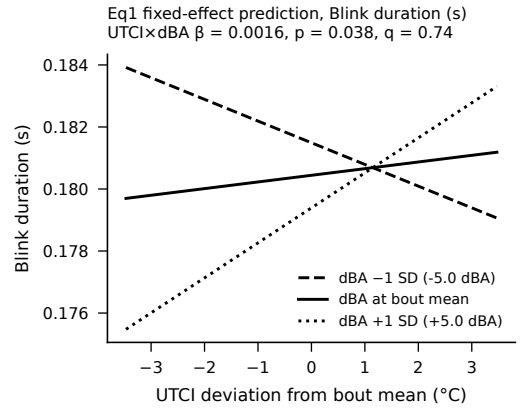
b

UTCI × sound in Eq1 (22 measures each replace dBA).
Cells: min raw p across 26 outcomes. BH q < 0.05: 0 of 572
in every run; min q per run: 1.00, 0.86, 0.98, 0.84, 0.98, 0.97

	10 s	30 s	60 s	300 s	10 s, lag 30 s	10 s, lag 60 s
dominant_frequency_Hz	0.063	0.086	0.105	0.075	0.073	0.168
freq_proportion_2000_to_20000Hz	0.173	0.077	0.076	0.053	0.066	0.082
freq_proportion_200_to_2000Hz	0.038	0.026	0.075	0.043	0.041	0.104
freq_proportion_20_to_200Hz	0.048	0.044	0.067	0.050	0.063	0.114
mechanisms_proportion	0.031	0.093	0.082	0.034	0.053	0.168
octave_proportion_1000_0_Hz	0.053	0.011	0.077	0.126	0.109	0.132
octave_proportion_125_0_Hz	0.052	0.126	0.015	0.036	0.027	0.080
octave_proportion_2000_0_Hz	0.158	0.030	0.056	0.110	0.043	0.142
octave_proportion_250_0_Hz	0.022	0.036	0.038	0.010	0.079	0.057
octave_proportion_31_5_Hz	0.010	0.063	0.073	0.078	0.036	0.053
octave_proportion_4000_0_Hz	0.104	0.048	0.184	0.054	0.113	0.060
octave_proportion_500_0_Hz	0.114	0.039	0.109	0.041	0.034	0.234
octave_proportion_63_0_Hz	0.028	0.036	0.194	0.220	0.057	0.229
octave_proportion_8000_0_Hz	0.141	0.060	0.091	0.091	0.043	0.018
rural_natural_proportion	0.052	0.004	0.078	0.099	0.008	0.002
spectral_centroid_Hz	0.125	0.057	0.080	0.047	0.109	0.126
spectral_rolloff85_Hz	0.042	0.007	0.052	0.061	0.039	0.082
urban_manmade_negative_proportion	0.059	0.016	0.107	0.155	0.107	0.078
urban_manmade_neutral_proportion	0.031	0.023	0.007	0.070	0.119	0.084
urban_manmade_positive_proportion	0.019	0.065	0.061	0.024	0.059	0.077
vehicle_proportion	0.053	0.031	0.141	0.061	0.020	0.100
wind_proportion	0.014	0.003	0.081	0.022	0.006	0.032

(b, cont.) UTCI × dBA in the primary family (26 outcomes):
Eq1 10 s (primary): min q = 0.74; q<0.05 in 0/26
Eq1 exposures lag 60 s: min q = 0.14; q<0.05 in 0/26
Eq1 phase means: min q = 0.99; q<0.05 in 0/26

c



Exported Eq1 fixed effects (not a refit); covariates at bout mean, phase terms 0 (mean of phases). 1 SD bout-centred deviation = 1.73 °C UTCI, 4.99 dBA. No CI: CR2 covariance not exported.